ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** sour	Acid.
2	**(d)** electricity	Electric energy.
3	**(d)** metal	Metal.
4	**(a)** physical change	Physical.
5	**(d)** sexually mature	Maturity.
6	**(d)** hot to cold	Always.
7	**(c)** second	Second.
8	**(a)** mouth	Ingestion.
9	**(c)** photosynthesis	Using sunlight.
10	**(a)** straight line	Straight.
11	**(c)** axis	Axis.
12	**(d)** hydrochloric acid	HCl.
13	**(a)** copper	Good conductor.
14	**(a)** copper	Metal.
15	**(b)** melting of ice	Melting.
16	**(b)** boys	Puberty.
17	**(c)** sinks	Heavier.
18	**(c)** distance / time	Speed formula.
19	**(c)** lungs	Lungs.
20	**(d)** transpiration	Evaporation.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	A sour-tasting substance (1); Vinegar (1).	1+1
22	A source of electricity (1); It has chemicals (1).	1+1
23	Iron (1); Copper (1).	1+1
24	No new substance (1); E.g. melting (1).	1+1
25	Time of sexual maturity (1); Between 10 and 14 years (1).	1+1
26	Conduction (1); Convection and radiation (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Lemon (1); Vinegar (1); Curd (1).	1+1+1
28	Cell (1); Bulb (1); Switch (1).	1+1+1
29	Lustre (1); Malleable (1); Good conductors (1).	1+1+1
30	Melting of ice (1); Boiling of water (1); Cutting paper (1).	1+1+1
31	Height increases (1); Voice deepens in boys (1); Body hair grows (1).	1+1+1
	OR: Spoon in hot tea (1); Iron rod in fire (1); Hot pan handle (1).	1+1+1
32	Second (1); Minute (1); Hour (1).	1+1+1
	OR: Mouth (1); Stomach (1); Intestine (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Acid sour, base bitter (1); Acid: blue litmus red (1); Base: red litmus blue (1); Base feels soapy (1); Examples differ (1).	1+1+1+1+1
34	Take a cell, a bulb and wires (1); Join one end of the bulb to + of the cell (1); Join the other end to - of the cell (1); The circuit is complete (1); The bulb glows (1).	1+1+1+1+1
35	Lustre (1); Malleability (1); Ductility (1); Conductivity (1); Sonority (1).	1+1+1+1+1
36	Melting (1); Boiling (1); Freezing (1); Tearing paper (1); Dissolving sugar (1).	1+1+1+1+1
	OR: Height (1); Voice (1); Hair (1); Sweat glands (1); Mood (1).	1+1+1+1+1
37	Heat moves in a solid (1); Spoon in hot tea (1); Particle to particle (1); Metals conduct (1); Wood does not (1).	1+1+1+1+1
	OR: Speed = distance / time (1); 120/20 (1); 6 (1); m/s (1); 6 m/s (1).	1+1+1+1+1
38	Ingestion (1); Digestion (1); Absorption (1); Assimilation (1); Egestion (1).	1+1+1+1+1
	OR: Occurs in leaves (1); Needs light (1); Needs CO_2 and water (1); Makes glucose (1); Gives oxygen (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** image	Reflection.
2	**(c)** hot	Warm.
3	**(a)** vinegar	Vinegar.
4	**(a)** cap	Metal cap.
5	**(c)** metal oxide	Oxide.
6	**(c)** carbon dioxide	Fizz.
7	**(d)** parents and teachers	Trusted adults.
8	**(c)** solid	Solid.
9	**(c)** uniform	Uniform.
10	**(c)** heart	Heart.
11	**(c)** green	Colour.
12	**(a)** reflection	Equal.
13	**(c)** Sun and Earth	Blocks Sun.
14	**(b)** soapy	Bases are bitter.
15	**(b)** two or more cells	Combination of cells.
16	**(a)** non-metal	Non-metal.
17	**(c)** chemical change	Chemical.
18	**(c)** 11 and 19 years	Teen.
19	**(b)** radiation	Radiation.
20	**(c)** 60 seconds	60 s.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Distance per unit time (1); Unit is m/s (1).	1+1
22	Breaking food into simple parts (1); In the alimentary canal (1).	1+1
23	Food making by plants (1); Needs sunlight (1).	1+1
24	Gives its own light (1); E.g. the Sun (1).	1+1
25	Spinning on an axis (1); Causes day and night (1).	1+1
26	Turns red (1); Shows acid (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Carbon dioxide + water (1); Sunlight and chlorophyll (1); Glucose + oxygen (1).	1+1+1
28	Sun (1); Candle (1); Bulb (1).	1+1+1
29	Day and night (1); Sunrise and sunset (1); Apparent motion of stars (1).	1+1+1
30	Citric: lemon (1); Lactic: curd (1); Acetic: vinegar (1).	1+1+1
31	Cell: long and short lines (1); Bulb: circle with a cross (1); Switch: a gap with a lever (1).	1+1+1
	OR: Good conductor (1); Ductile (1); Long lasting (1).	1+1+1
32	Rain (1); Seasons (1); Day and night (1).	1+1+1
	OR: Tobacco (1); Alcohol (1); Drugs (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Luminous give light (1); Non-luminous do not (1); Sun vs Moon (1); Seen by own light vs reflected (1); Examples (1).	1+1+1+1+1
34	Earth is a sphere (1); Spins on its axis (1); Faces the Sun: day (1); Away: night (1); 24 hours cycle (1).	1+1+1+1+1
35	Litmus: red, blue (1); Turmeric: yellow, red (1); China rose: pink, green (1); Acid first (1); Base second (1).	1+1+1+1+1
36	Light (1); Heat (1); Fan (1); TV (1); Cooking (1).	1+1+1+1+1
	OR: Burn sulphur (1); Gas forms (1); Dissolve in water (1); Acid forms (1); Blue litmus turns red (1).	1+1+1+1+1
37	Add alum to hot water (1); Stir (1); Filter (1); Cool (1); Crystals form (1).	1+1+1+1+1
	OR: Chemical messengers (1); Made by glands (1); Travel in blood (1); Control growth (1); Control mood (1).	1+1+1+1+1
38	Hot tea (1); Cold drinks (1); Soup (1); Travel (1); Picnics (1).	1+1+1+1+1
	OR: 18 x 5/18 (1); 5 m/s (1); 10 x 18/5 (1); 36 km/h (1); Use the factor (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** teeth	Teeth.
2	**(a)** chlorophyll	Pigment.
3	**(c)** luminous object	Gives light.
4	**(d)** Sun	Sun.
5	**(a)** sulphuric acid	It has sulphur.
6	**(b)** plastic	Insulation.
7	**(d)** rubber	Non-metal.
8	**(b)** burning of paper	Burning.
9	**(b)** girls and women	Females.
10	**(d)** in the day	From sea to land.
11	**(a)** m/s	Metre per second.
12	**(d)** nose	Nostrils.
13	**(a)** insects	Insectivorous.
14	**(c)** reflection	Reflection.
15	**(b)** cold	Cool.
16	**(b)** soap	Soap.
17	**(c)** flat base	Flat.
18	**(b)** non-metal oxide	Oxide.
19	**(c)** carbon dioxide	CO_2.
20	**(d)** friends	Same age.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Copper (1); Iron (1).	1+1
22	Drawn into wires (1); Copper wire (1).	1+1
23	Iron oxide (1); Reddish-brown (1).	1+1
24	Thyroid (1); Makes thyroxine (1).	1+1
25	Iron (1); Copper (1).	1+1
26	Time for one oscillation (1); In seconds (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Cotton shirt (1); Light colours (1); Loose clothes (1).	1+1+1
28	Distance = speed x time (1); 40 x 2 (1); 80 km (1).	1+1+1
29	Amylase (1); Pepsin (1); Lipase (1).	1+1+1
30	Anchor plant (1); Absorb water (1); Absorb minerals (1).	1+1+1
31	Erect (1); Same size (1); Virtual (1).	1+1+1
	OR: A star (1); Source of light (1); Hot (1).	1+1+1
32	Turns red (1); Soap is basic (1); Yellow in acid (1).	1+1+1
	OR: To turn on (1); To turn off (1); To save energy (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Two kidneys (1); Bean shaped (1); Filter blood (1); Make urine (1); Remove waste (1).	1+1+1+1+1
34	Trap insects (1); Pitcher plant (1); Venus flytrap (1); Need nitrogen (1); Have green leaves (1).	1+1+1+1+1
35	Three cards (1); Holes aligned (1); Candle on one side (1); See the light (1); Shift one card: no light (1).	1+1+1+1+1
36	Full moon (1); Earth between (1); Shadow on Moon (1); Moon looks dark red (1); Safe to watch (1).	1+1+1+1+1
	OR: Take red and blue litmus (1); Dip in the solution (1); See the colour change (1); Acid turns blue to red (1); Base turns red to blue (1).	1+1+1+1+1
37	Copper (1); Iron (1); Aluminium (1); Rubber (1); Plastic (1).	1+1+1+1+1
	OR: Lustre (1); Hardness (1); Brittleness (1); Conduction (1); Sound (1).	1+1+1+1+1
38	Burning (1); Rusting (1); Cooking (1); Digestion (1); Fermentation (1).	1+1+1+1+1
	OR: Pituitary: growth hormone (1); Thyroid: thyroxine (1); Pancreas: insulin (1); Testes: testosterone (1); Ovary: oestrogen (1).	1+1+1+1+1

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